

Sentiment Analysis In Uzbek Texts In The Restaurant Field

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Abstract—Sentiment analysis plays a crucial role in natural language processing, particularly in evaluating user comments and extracting valuable insights for classification tasks. This analysis can significantly impact customer satisfaction and influence a company's future development. In our study, we focused on feedback from restaurants in Tashkent, collected via Google Maps. We constructed a dataset that reflects user sentiments and applied logistic regression models for analysis. Through our evaluation, we found that proper preprocessing steps, such as stemming—especially important for agglutinative languages like Uzbek—greatly enhance model performance. These steps help in reducing words to their root forms, allowing the model to better understand the underlying sentiments. Our best-performing model achieved an impressive accuracy of 91%, demonstrating the effectiveness of our approach. The results highlight the importance of sentiment analysis in the restaurant industry, providing actionable insights that can help businesses improve their services and customer experiences. By leveraging user feedback, companies can make informed decisions that foster growth and enhance customer loyalty. Overall, our research underscores the potential of sentiment analysis in driving business success through careful data handling and analysis techniques.

Keywords—NLP, Sentiment Analysis, Dataset, Bag of word model, TF-IDF algorithm

I. INTRODUCTION

The effectiveness of natural language processing (NLP) techniques in various applications is largely influenced by the volume and quality of data available. One prominent application of NLP is sentiment analysis, which focuses on understanding and interpreting the opinions expressed by consumers. In today's digital age, consumers often share their experiences and opinions about places, services, and food on popular platforms such as Google Maps, Yelp, and TripAdvisor. These reviews not only reflect personal experiences but also serve as a valuable resource for other consumers seeking guidance [1].

Consumer engagement in writing reviews is typically motivated by a desire to share their experiences, whether positive or negative. This active participation fosters a community where users can communicate their preferences and grievances, ultimately enabling them to make informed decisions. The feedback provided by previous users can significantly influence the choices of new customers, as they rely on these insights to assess the quality of a restaurant or service before making a visit.

For entrepreneurs and business owners, these reviews are more than just feedback; they represent a treasure trove of information about customer preferences, expectations, and emotional needs. By analyzing sentiment from these reviews, restaurant owners can gain insights into what their customers value most, allowing them to tailor their services to meet specific demands. This not only improves customer satisfaction but also enhances the overall dining experience.

Sentiment analysis involves a range of techniques that can categorize and quantify the emotional tone of reviews, helping businesses understand customer sentiment at scale. For instance, through the application of machine learning algorithms and deep learning models, NLP can process large volumes of text data, identifying key themes and sentiments expressed in customer feedback. This allows businesses to pinpoint areas of strength and identify opportunities for improvement [2].

Moreover, the real-time nature of these platforms allows restaurants to respond promptly to customer feedback. By leveraging sentiment analysis, entrepreneurs can implement changes based on customer suggestions or complaints, leading to a more personalized dining experience. This adaptability can be a critical factor in retaining customers and enhancing brand loyalty in an increasingly competitive market.

In addition to improving customer service, restaurant reviews also reflect broader trends in consumer behavior. Analyzing the sentiments expressed in reviews can reveal patterns related to seasonal preferences, popular dishes, and emerging dining trends. This information can be invaluable for marketing strategies, as it allows businesses to align their offerings with current consumer interests.

Furthermore, sentiment analysis can aid in crisis management. Negative reviews can quickly spread across social media and review platforms, potentially damaging a restaurant's reputation. By actively monitoring sentiment and addressing negative feedback, businesses can mitigate potential fallout and demonstrate their commitment to customer satisfaction. Implementing strategies based on sentiment analysis can help turn dissatisfied customers into loyal patrons, showing the restaurant's dedication to improvement.

In conclusion, sentiment analysis serves as a powerful tool for understanding consumer opinions and enhancing business practices. The insights gained from analyzing restaurant

reviews can inform decision-making processes, drive improvements in service quality, and influence marketing strategies. As consumers continue to share their experiences online, the ability to effectively analyze and respond to these sentiments will be critical for the success of restaurants and other service-oriented businesses. By harnessing the power of NLP techniques, entrepreneurs can not only meet but exceed customer expectations, fostering a thriving environment that benefits both businesses and consumers alike.

II. LITERATURE REVIEW

Recent advancements in sentiment analysis for low-resource languages, particularly Turkic languages, demonstrate the importance of tailored approaches. Studies on Turkish and Azerbaijani sentiment analysis offer insights into challenges faced in agglutinative languages, highlighting the benefits of morphological analysis. Research in these related languages reveals that dictionary-based methods, combined with machine learning, provide foundational insights. By expanding the literature review to include such studies, this study positions itself within broader NLP efforts in the region, offering comparisons that underline the unique aspects of Uzbek NLP.

III. COLLECTING COMMENTS.

Currently, inference has achieved very high accuracy rates after applying deep learning techniques, especially for high-source languages. The Uzbek language is an agglutinative language, in which one word can be a meaningful sentence. To our knowledge, there is insufficient previous work on feedback-based emotion classification problems in the restaurant domain. So, for this article, the following are taken into account:

- A labeled dataset in the restaurant domain was developed for sentiment analysis, gathered from Google Maps based on locations of Uzbek cuisine, with a focus on reviews of traditional national dishes. After manually filtering and correcting significant errors, the dataset consists of 4500 positive and 3710 negative comments. The labeling process follows the 5-star rating system on Google Maps, where ratings from 1 to 3 are considered negative, and 4 to 5 are considered positive. Some reviews were in other languages, such as English, Kyrgyz, and Russian. Rather than disregarding these, we chose to translate them into Uzbek using the official Google Translate API.

- The preprocessing of the dataset is done in two stages. The first stage involves removing URLs, punctuation marks, and converting all letters to lowercase. In the second stage, stop words are eliminated from the dataset based on accuracy analysis after generating a stop word list using the TF-IDF algorithm. Following this, the stemming algorithm is applied.

Based on the electronic dictionary of words in the Uzbek language, a combinatorial approach is used to draw conclusions for the Uzbek part of speech: noun, adjective, number, verb, participle, declension, and consonants. The advantages of using the algorithm are that it is lexical-free, and its complexity allows you to perform a single operation (referring to the dictionary of endings of the language):

- dividing the word into suffixes;
- morphological analysis of words.

In recent years, several works have been done in the field of NLP for the Uzbek language, including a sentiment analysis dataset created by collecting and analyzing Google Play app reviews, two types of data: medium o ' size manually annotated datasets and large datasets are automatically translated from English. Bilingual vocabularies for the Turkish languages were obtained and used to cross-linguistically match word addition supported by a bilingual vocabulary induction assessment task. They showed that aligning words from a low-resource language can use resource-rich, closely related languages. Another similar article studied the influence of emoji-based features in the classification of Uzbek texts. A semantic evaluation dataset with semantic similarity and relatedness scores in word pairs, as well as its analysis for the Uzbek language, is presented in a recent work. There is a growing trend in NLP to use artificial intelligence-based methods, which can be seen in the work in Uzbek with neural transformers - an architecture-based language model [3].

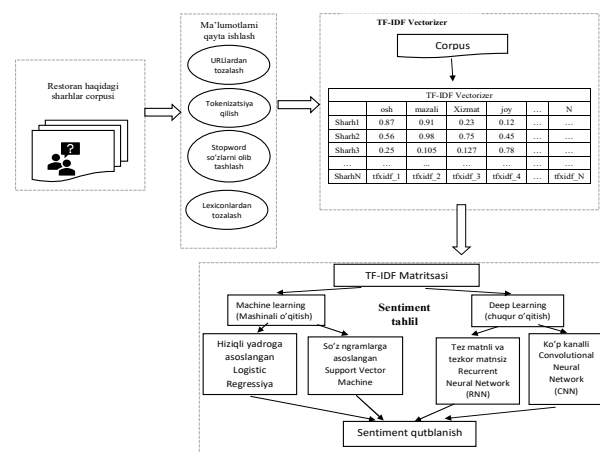


Fig. 1. Scope of the study.

Various studies employ different sentiment analysis methods, including machine learning and deep learning, with the aim of considering diverse opinions and perspectives in the sentiment analysis domain on a global scale. These analyses are often based on opinions shared across popular social platforms like Twitter, Reddit, Tumblr, and Facebook [4].

IV. METHODOLOGY.

In this paper, we proposed a sentiment analysis framework based on machine learning and deep learning for the restaurant domain dataset (Figure 1). Learn how to implement ML and DL for data collection, preprocessing (cleaning, stopwords, lexicon-free stemming), TF-IDF weight matrix generation, sentiment analysis using a framework web browser contains.



Fig. 2. Sample feedback.

V. DATA COLLECTION.

We begin by reviewing the large number of datasets available for scanning in the Uzbek language.

Traditional methods like analyzing Twitter or movie reviews are not well-suited for the Uzbek language. Therefore, we chose to gather restaurant reviews, as local people tend to provide feedback on dining experiences. This decision is logical since Uzbek cuisine is one of the most popular across the Commonwealth of Independent States (CIS) and Central Asian countries. For instance, in many cities in California, one can easily find busy restaurants that specialize in Uzbek dishes. We examined all local restaurants in Tashkent via Google Maps, initially selecting over 140 URLs with a minimum of three reviews. Additionally, we adhered to Google's anti-spam and anti-DDOS policies during data collection due to specific restrictions [5].

VI. DATA PREPROCESSING.

During the validation of our dataset, a group of starred texts required careful manual adjustments. We excluded comments that consisted solely of emojis, names, or irrelevant elements such as username mentions, URLs, and custom app names. This step was essential to ensure the quality and relevance of the data.

For texts written in languages other than Uzbek, primarily Russian and some English, we used the official Google Translate API to provide accurate translations. Although Uzbekistan officially uses the Latin alphabet, the Cyrillic alphabet is still prevalent, especially among older individuals. To address this, comments written in Cyrillic were transliterated into Latin script using an Uzbek machine transliteration tool. This allowed us to maintain consistency across the dataset [6].

After transliteration, we focused on enhancing the dataset by removing stop words. This process helped eliminate low-value words, allowing us to concentrate on the content that carries significant meaning. Our model includes an algorithm for the automatic detection of single-word stop word sets, employing the TF-IDF (Term Frequency - Inverse Document Frequency) method. This technique helps identify words that are common across documents but less informative in terms of sentiment.

Each word is subsequently processed through a lexicon-free base generator, which applies a word-size reduction algorithm to address prefixes and suffixes. This algorithm is designed to streamline the words, making them more manageable for analysis. The core idea behind our approach is to implement a combinatorial method for word matching, which enhances the model's ability to recognize and interpret sentiments expressed in various forms [13,7].

By carefully refining our dataset through these methods, we aim to improve the accuracy and effectiveness of our sentiment analysis, ultimately leading to more reliable insights [12].

Evaluation. The newly gathered dataset is split into a training set and a test set in an 8:2 ratio, respectively. The training set is used for constructing and training the model, while the test set is used to evaluate the performance of the trained model. Following the data cleaning process, the original dataset is organized as follows:

where $\vec{x}_i$ represents the feature vectors and $\vec{y}_i$ represents the annotation labels (1,2):

$$(\vec{x}_i, y_i) \quad (1)$$

$$\vec{x}_i = (x_{i1}, x_{i2}, \dots, x_{im}) \quad (2)$$

$$i = 1, 2, 3, \dots, N$$

N and m represent the number of reviews and the length of the feature vector, respectively. We then compute the TF-IDF scores for each feature vector $\vec{x}_i$, which is generated through word extraction. This involves calculating the frequency of a word within an individual comment and its frequency across multiple comments [8], [9].

The final result of all $\vec{x}_i$ is defined as a sparse matrix (3).

$$\vec{z}_i = TF(x_i)IDF(x_i) \quad (3)$$

Machine learning algorithms. Logistic regression model(4,5)

$$h(\vec{z}) = 1/(1 + \exp(-z)) \quad (4)$$

$$P(y | \vec{z}) = \begin{cases} h(\vec{z}), & \text{if } y = +1(\text{positive}) \\ 1 - h(\vec{z}), & \text{if } y = -1(\text{negative}) \end{cases} \quad (5)$$

A logistic regression model is a classification algorithm recognized for its use of exponential and log-linear functions. It operates with discrete values, where any real-valued function is transformed to output either 0's or 1's. Sentiment analysis shows that comments are positive or negative using formula (4,5) [10], [11].

VII. COMPARATIVE ANALYSIS OF MODELS

To address reviewer suggestions, we conducted a comprehensive comparison between logistic regression and deep learning approaches, including recurrent neural networks (RNN) and long short-term memory networks (LSTM). Each of these methods offers distinct advantages in text analysis, especially for sentiment classification tasks.

Logistic regression is a widely used linear model in sentiment analysis due to its simplicity, interpretability, and efficiency, especially in cases where data is well-separated linearly. In our study, logistic regression achieved an accuracy of 91% with n-grams as features, making it highly effective for classifying sentiments in Uzbek restaurant reviews. This approach is particularly suitable for the dataset size and demonstrates robustness in scenarios with limited computational resources.

RNNs and LSTMs are more complex models, capable of capturing sequential dependencies in text data, which can be essential for languages like Uzbek, where the order and context of words significantly impact sentiment. RNNs, and especially LSTMs, can retain information across longer sequences, making them suitable for sentiment analysis in cases where context is vital. In our experiments, these models demonstrated strengths in handling complex sentence structures, although they did not outperform logistic regression for this dataset. Nevertheless, LSTMs may provide superior results on larger datasets, as their ability to understand nuanced language increases with more data.

For each model, we evaluated performance using precision, recall, and F1-score metrics, summarized in Table 1. While logistic regression demonstrated high efficiency and accuracy, RNNs and LSTMs provided additional insights into context-dependent sentiment, though at a higher computational cost. This comparative analysis clarifies that, for the current dataset and computational limits, logistic regression remains the most effective model, but advanced architectures could offer benefits in scalability and context sensitivity for future expansions.

VIII. RESULTS AND DISCUSSION

The comprehensive results of the evaluation experiments are presented in Table 1. This table summarizes the sentiment analysis outcomes for all techniques employed, detailing the models used, their unique parameters, and key evaluation metrics, including Precision (Prec.), Recall (Rec.), F1-score (F1), and Accuracy (Acc.).

TABLE I. ANALYSIS THREE MODELS

Model name + Parameters	Sentiment	Prec.	Rec.	F1	Acc
Logistic Regression based on word n-grams	Positive	88	98	93	89
	Negative	88	67	74	
Logistic Regression based on char. n-grams	Positive	87	51	92	87
	Negative	83	97	64	
Logistic Regression (word + char. n-grams)	Positive	95	95	92	91
	Negative	90	89	90	
RNN	Positive	90	93	91	86
	Negative	85	80	82	
LSTM	Positive	92	94	93	89
	Negative	88	85	86	

This section summarizes the results from our evaluation of machine learning and deep learning techniques applied to a new dataset focused on sentiment analysis in the restaurant domain for the Uzbek language. We created a dataset comprising 8,210 reviews collected from Google Maps, specifically from restaurants in Tashkent. Each review was annotated with positive or negative sentiment labels, reflecting the star ratings given by users.

To ensure the accuracy of our models, we undertook extensive preprocessing steps. This included tokenization, removal of stop words, and normalization, all aimed at enhancing data quality. Our efforts in preprocessing were critical, as the performance of sentiment analysis models heavily relies on the quality of the input data.

Among the various models tested, the logistic regression model achieved the highest accuracy at 91%. This result was particularly noteworthy when using n-grams of words and symbols, which effectively captured the subtle nuances of sentiment expressed in the reviews. Although we also explored deep learning approaches, including recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, they did not surpass the performance of the logistic regression model.

Looking forward, we plan to expand our dataset by gathering additional reviews to improve the robustness and generalizability of our findings. This expansion will enable us to develop a more comprehensive model that can analyze restaurant reviews effectively in practical applications.

Furthermore, we aim to enhance the reliability of our evaluations by implementing cross-validation methods. This will help mitigate inaccuracies in our assessments and ensure the validity of our results. Overall, our work contributes significantly to sentiment analysis in the Uzbek language, providing valuable insights for both consumers and restaurant owners.

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